

Physics-informed hybrid U-Net architecture for phase-to-height reconstruction in dual divergent structured-light systems

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ABSTRACT

Three-dimensional surface reconstruction from structured-light phase measurements in dual-projection divergent systems presents challenges due to nonlinear phase–height relationships, spatially variant sensitivity, and depth-dependent geometric distortions. This work introduces PIHUNN v4.2, a physics-informed hybrid U-Net architecture designed to learn the inverse mapping from wrapped phase ϕ_w to surface height h while respecting the physical constraints imposed by divergent illumination geometry. The proposed method integrates multi-scale feature extraction through a Feature Descriptor with Different Scales (FDDS) module, residual pre-activation blocks for stable gradient propagation, and Squeeze-and-Excitation (SE) attention for channel recalibration. A hybrid loss function combining mean squared error, structural similarity, gradient fidelity, multi-scale gradient consistency, total variation regularization, and height magnitude penalty enforces photometric and geometric coherence. The gradient terms are normalized by the physical bound $2\pi/\lambda_{\min}$ derived from the minimum effective wavelength, ensuring that predicted surfaces do not violate optical constraints. Experimental validation on a dataset of 4000 synthetic phase–height pairs demonstrates that PIHUNN v4.2 achieves RMSE of 0.0565 ± 0.0243 , SSIM of 0.996 ± 0.004 , and PSNR of 25.6 ± 3.1 dB, outperforming polynomial calibration, stereo-triangulation, and state-of-the-art deep learning baselines including U-Net and U-Net + +. Qualitative validation on real stereomoiré measurements confirms the generalization capability of the network to experimental conditions. The physics-informed constraints provide robustness in high-gradient regions where classical methods fail, enabling accurate reconstruction across the full measurement volume of divergent dual-projection systems.

1. Introduction

Three-dimensional (3D) surface reconstruction from structured-light phase measurements is a fundamental task in optical metrology, particularly in interferometry, profilometry, and stereomoiré systems [1–3]. Conventional phase-to-height calibration relies on geometric models that assume telecentric projection, linear phase-height response, and symmetric epipolar geometry [4]. These assumptions fail in practical systems that employ divergent illumination or dual-projection configurations, which introduce nonlinear distortions in the wrapped phase ϕ_w and degrade the geometric consistency of the reconstructed height map h .

In divergent structured-light systems, the mapping

$$\phi_w \longrightarrow h$$

becomes spatially variant due to non-uniform sensitivity, asymmetric projection rays, and depth-dependent scale factors [5,6]. Moreover, dual-projection arrangements exacerbate these nonlinearities: each projector contributes a distinct phase-geometry relationship, making analytical

calibration unstable or sensitive to noise. As a result, classical polynomial regression, stereo-triangulation formulations, and least-squares fitting often fail to maintain accuracy outside narrow calibration volumes [7] which provides the experimental and geometrical foundation for the dual-divergent projection configuration adopted in this study.

Recent advances in deep learning have significantly impacted optical metrology [8], including phase unwrapping [9,10] and structured-light reconstruction under geometric constraints [11]. However, most architectures rely solely on statistical feature extraction (e.g., U-Net, ResNet, MA-Net), lacking explicit physical priors. These networks tend to oversmooth high-gradient regions, lose topological coherence under changes in illumination or sensitivity, and generalize poorly when the phase-height relation is non-monotonic. Physics-informed neural networks (PINNs) partially address these issues by embedding physical constraints into the loss function [12–14], yet PINNs alone struggle with high-dimensional image-to-image regression.

To overcome these limitations, we propose a physics-informed hybrid U-Net (PIHUNN v4.2) designed specifically for dual-projection divergent

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structured-light systems. Unlike purely data-driven approaches such as PhaseNet [15,16], the method integrates: (i) multi-scale feature extraction through a Feature Descriptor with Different Scales (FDDS) module; (ii) residual pre-activation blocks for stable deep representation; (iii) Squeeze-and-Excitation (SE) attention for channel recalibration; and (iv) a hybrid loss formulation that enforces photometric fidelity, geometric consistency, gradient coherence, and total-variation regularity. The physical gradient bound derived in Section 3 ensures that the predicted surface $\hat{h}$ respects the maximum allowable phase gradient determined by the minimal effective wavelength.

The contributions of this work are threefold:

1. We introduce a physics-informed architecture tailored to divergent dual-projection systems, incorporating geometric constraints into the learning process.
2. We develop a hybrid perceptual-physical loss (MSE + SSIM + gradient fidelity + multi-scale gradients + TV + L_2 height penalty) that preserves fine structures while maintaining stability under noise and configuration changes.
3. We demonstrate that PIHUNN v4.2 outperforms classical calibration models and deep learning baselines, improving RMSE, SSIM, and PSNR across all evaluated datasets. These results update and extend the comparative analysis in Table 2.
4. We also present a robustness study showing that the physically normalised gradient term stabilises the reconstruction when the fringe period varies by a factor of three without retraining the network

Unlike prior learning-based approaches that primarily address phase unwrapping or denoising, the proposed PIHUNN framework targets the inverse phase-to-height mapping under strongly divergent projection, where analytical calibration becomes highly nonlinear and spatially variant. By embedding physically motivated gradient constraints into the learning objective, PIHUNN effectively replaces explicit phase-to-height calibration with a data-efficient, physics-consistent regularization strategy. This enables robust surface reconstruction across varying depths and geometries without requiring per-setup recalibration.

The remainder of the manuscript is organized as follows. Section 2 reviews related work in phase-based metrology and learning-based reconstruction. Section 3 formulates the nonlinear phase–height relationship in divergent dual-projection systems. Section 4 describes the proposed PIHUNN architecture and the physics-informed hybrid loss. Section 5 details the dataset and training protocol. Section 6 presents quantitative and qualitative results. Section 7 discusses application scenarios and practical considerations. Finally, Section 8 concludes the paper.

2. Related work

Research on phase-based 3D reconstruction has followed two major directions: analytical calibration methods derived from optical geometry, and learning-based approaches designed to approximate the nonlinear mapping from wrapped phase ϕ_w to height h . Early structured-light and interferometric models [5,6] assume telecentric projection, uniform sensitivity, and symmetric ray geometry, conditions that simplify the triangulation equations and yield quasi-linear mappings between phase and surface depth [17,18]. In practical systems employing divergent illumination, these assumptions break down and the phase-height response becomes spatially variant, depth dependent, and sensitive to small perturbations in alignment, fringe frequency, or projector-camera orientation.

Dual-projection systems magnify these effects because each projector introduces a distinct geometric sensitivity, creating two heterogeneous phase-height relationships that cannot be simultaneously linearized. Works such as [7,19] demonstrate that polynomial fitting and regression-based calibration improve accuracy within restricted depth intervals but remain unstable when the scene extends beyond the calibrated volume or when the system configuration changes. These limitations highlight the structural fragility of purely analytical calibration in non-telecentric, multi-source projection scenarios Fig. 1.

Collimated and divergent illumination lead to fundamentally different phase–geometry relationships in dual-projection structured-light systems, which directly affects the conditioning of the inverse problem. In collimated projection, the fringe period remains constant and the moiré spacing is spatially uniform, simplifying the phase-to-height mapping but restricting the measurable depth range. In contrast, divergent projection inherently introduces a position-dependent period and a non-uniform moiré spacing $d_R(x, z)$, producing a nonlinear and depth-coupled phase response. Although this configuration significantly increases sensitivity and extends the usable measurement volume, it also amplifies geometric distortions and breaks the linear calibration models traditionally used. This motivates the need for physics-informed learning approaches such as PIHUNN, capable of absorbing the nonlinearity introduced by the divergent geometry and maintaining robustness across varying configurations.

Deep learning architectures have gained attention as an alternative path to learning nonlinear phase-height mappings directly from data. U-Net variants [20], encoder-decoder backbones [21], and attention-based models have achieved competitive performance in fringe analysis, phase unwrapping, and dense depth regression [8,15,22–24]. Their multi-scale feature extraction and skip connections enable preservation of low-frequency shape information while capturing local surface variations. Nonetheless, CNNs trained solely with statistical losses such as MSE or SSIM often oversmooth high-gradient regions, lose topological coherence under noise, and generalize poorly when the optical configuration varies [25]. Because these architectures have no knowledge of the geometric sensitivity or interferometric constraints of the system, the predicted height maps may violate physical limits on allowable gradients, exhibit depth-dependent biases, or drift when the phase-height relationship becomes non-monotonic.

Physics-informed learning frameworks, particularly physics-informed neural networks (PINNs), introduce physical priors directly into the optimization process by penalizing violations of governing equations or geometric constraints [12,13,26]. While powerful in low-dimensional problems, classical PINNs become computationally prohibitive for high-resolution image-to-image regression. Recent hybrid approaches combine CNN-based feature extraction with physically motivated regularization terms [27,28], achieving more stable behavior without the training cost of a full PINN. In optical metrology, these hybrid models incorporate gradient consistency, multi-scale smoothness, and bounds derived from the system sensitivity [29,30], providing better geometric stability compared to purely data-driven methods.

Within this context, the architecture proposed in this work (PIHUNN v4.2) follows the hybrid philosophy: multi-scale deep representations extracted by FDDS and residual pre-activation blocks are combined with physically grounded constraints that enforce coherence between the predicted surface $\hat{h}$ and the underlying phase-height sensitivity of divergent dual-projection systems (see Section 3). This integration addresses the two major weaknesses observed in prior approaches: (i) analytical calibration lacks flexibility under nonlinear divergence effects, and (ii) purely statistical CNNs lack physical interpretability and may violate geometric consistency. As demonstrated later in Table 2, this hybrid strategy enables PIHUNN v4.2 to outperform classical calibration models and state-of-the-art deep architectures across all evaluated metrics. As quantitatively analyzed in Section 6.3 and summarized in Table 4, the contribution of each architectural component is systematically evaluated.

3. Physical model of dual-divergent projection

Three-dimensional reconstruction from structured-light interferometric measurements relies on the forward mapping between surface height $h(x, y)$ and the observed wrapped phase $\phi_w(x, y)$. In dual-projection systems with divergent illumination, this mapping becomes nonlinear, spatially variant, and sensitive to geometric perturbations. Unlike telecentric or collimated arrangements, each projector emits rays with different divergence angles and baseline distances, producing a depth-dependent fringe period and a heterogeneous phase sensitivity across the field of

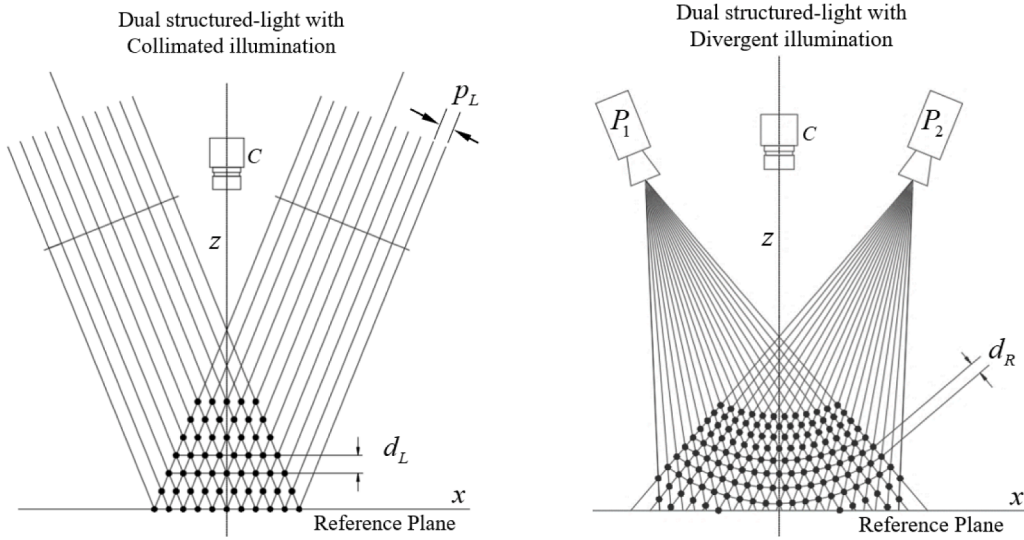


Fig. 1. Dual structured-light projection geometry. (a) Collimated illumination: the pattern period p_L is constant and the moiré fringe spacing d_L remains uniform across the entire field of view. (b) Divergent illumination: the pattern period varies with position and the moiré fringe spacing $d_R(x, z)$ depends on the local geometry. Adapted from [7].

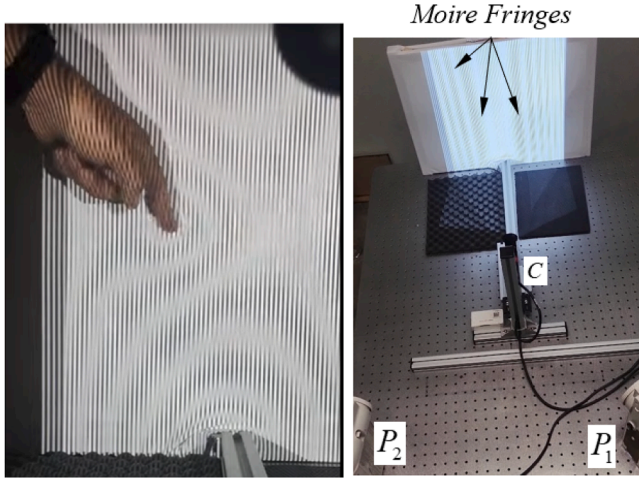


Fig. 2. Moiré fringe measurement system for surface deformation analysis. (a) Moiré fringe pattern deformation caused by an object (hand) placed between the gratings, (b) Experimental setup: P_1 and P_2 are projectors, C is the camera, and the screen displays the Moiré fringes.

view. This section formulates the forward model, establishes physically consistent bounds on the phase gradient, and highlights the geometric inconsistencies that motivate the hybrid learning approach introduced in Section 4.

Figs. 2 and 3 illustrates the configuration considered in this work. Two projectors P_1 and P_2 illuminate the object from distinct baselines and intersection angles, and the camera observes the resulting moiré patterns [31,32]. Because each projector has a different optical center and divergence angle, every pixel (x, y) is affected by a unique projection ray and therefore a unique local sensitivity.

3.1. Phase formation under divergent illumination

For each projector P_k ($k = 1, 2$), the observed fringe pattern at pixel (x, y) is modeled as [18]

$$I_k(x, y) = I_0(x, y) + I_m(x, y) \cos(\phi_k(x, y)), \quad (1)$$

where I_0 is the background illumination, I_m is the modulation amplitude, and $\phi_k(x, y)$ is the absolute phase induced by projector P_k . After demodulation, the wrapped phase becomes

$$\phi_{w,k}(x, y) = \text{mod}(\phi_k(x, y), 2\pi). \quad (2)$$

In divergent projection, the absolute phase depends on the effective spatial wavelength $\lambda_{\text{eff},k}(x, y)$:

$$\phi_k(x, y) = \frac{2\pi}{\lambda_{\text{eff},k}(x, y)} d_k(x, y), \quad (3)$$

where $d_k(x, y)$ is the optical path difference. The effective wavelength varies with position because local magnification and ray divergence modify the apparent fringe period:

$$\lambda_{\text{eff},k}(x, y) = \lambda_0 \frac{z(x, y)}{z(x, y) + \Delta_k(x, y)}, \quad (4)$$

with λ_0 the nominal projected period at the projector plane, and $\Delta_k(x, y)$ a geometric term capturing divergence and projector pose.

3.2. Nonlinear phase-height relationship

Let $h(x, y)$ denote the surface height relative to a reference plane. Under diverging projection, the forward model relating height to phase is

$$\phi_k(x, y) = f_k(h(x, y)), \quad (5)$$

where $f_k(\cdot)$ is a nonlinear, spatially varying function that depends on the projector-camera configuration, divergence angle, and depth.

The inverse problem is therefore

$$h(x, y) = f_k^{-1}(\phi_k(x, y)), \quad (6)$$

which is not analytically invertible and may become ill-conditioned when $\lambda_{\text{eff},k}$ varies rapidly or when the scene extends beyond the calibrated depth interval.

Dual projection imposes an additional geometric constraint. The surface must satisfy

$$f_1^{-1}(\phi_1(x, y)) = f_2^{-1}(\phi_2(x, y)) = h(x, y), \quad (7)$$

which is difficult to enforce directly when f_1 and f_2 have different sensitivities. In practice, noise, calibration drift, and diverging rays make (7) inconsistent, producing biased or unstable reconstructions.

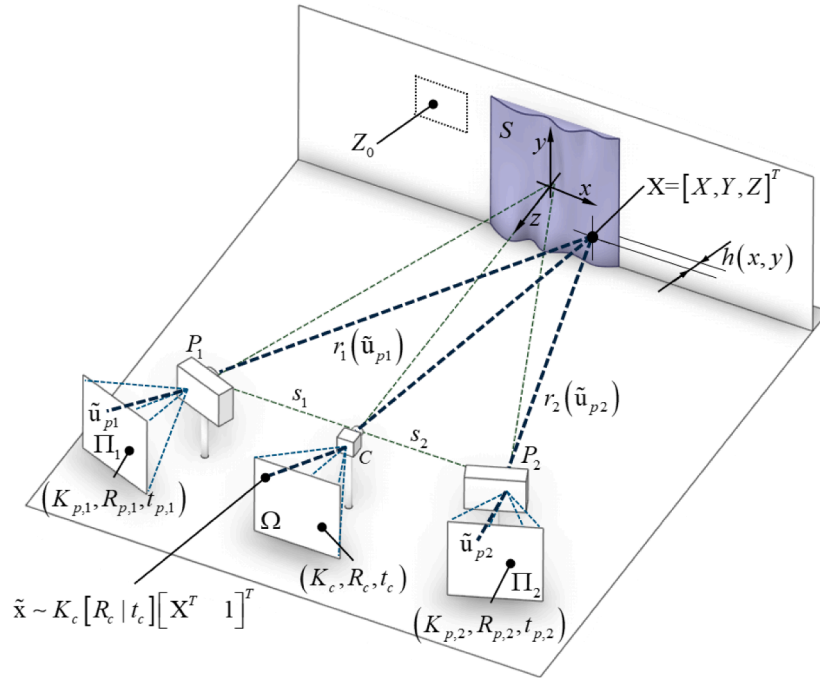


Fig. 3. Dual-divergent structured-light configuration. Each projector P_1 and P_2 emits non-telecentric, divergent rays, producing depth-dependent sensitivity and heterogeneous phase responses across the field of view.

3.3. Physical bound on the phase gradient

Differentiating (3), and assuming locally planar fringe fronts and first-order approximation, yields

$$\|\nabla \phi_k(x, y)\| \leq \frac{2\pi}{\lambda_{\text{eff},k}(x, y)}, \quad (8)$$

which expresses the fact that the maximum allowed phase gradient is governed by the local effective wavelength. As the surface approaches projector P_k , the effective wavelength decreases and the gradient increases.

The tightest global bound occurs at the minimum effective wavelength:

$$\|\nabla \phi_k(x, y)\| \leq \frac{2\pi}{\lambda_{\min}}, \quad (9)$$

where $\lambda_{\min}$ is determined by the closest depth and strongest local divergence.

The gradient term (Eq. (9)) is normalised by an upper bound on the admissible phase gradient. For the experimental stereo set-up used in this work, the moiré carrier has a minimum fringe period of approximately $\lambda_{\min}^{\text{train}} \simeq 16$ pixels on the camera sensor. This value is used as a fixed physical bound to normalise the gradient term during training. This constraint acts as a physical prior, preventing the network from predicting height gradients that contradict the optical limits of the system. In Section 6.4 we evaluate the trained networks on synthetic test sets with different fringe periods $\lambda_{\min}^{\text{test}}$, including values smaller than the experimental one, in order to stress the robustness of the model.

To connect this optical bound with the phase-to-height inverse problem, we recall from (5) that $\phi_k(x, y) = f_k(h(x, y))$. By the chain rule,

$$\nabla \phi_k(x, y) = \frac{\partial f_k}{\partial h}(x, y) \nabla h(x, y), \quad (10)$$

where $\partial f_k / \partial h$ represents the local phase-to-height sensitivity induced by the divergent projector-camera geometry. Therefore, bounding $\|\nabla \phi_k\|$ implies a physically meaningful scale for admissible height gradients. In practice, since $\partial f_k / \partial h$ varies spatially and may not be available in closed form for all configurations, we use the conservative global bound

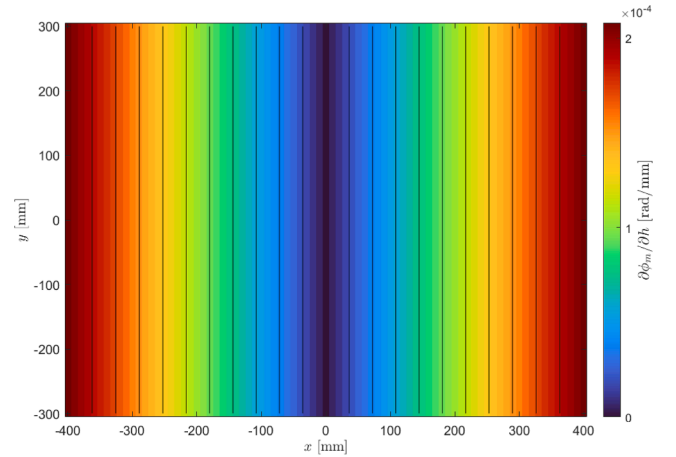


Fig. 4. Spatial distribution of the effective fringe wavelength $\lambda_{\text{eff}}(x, y)$ in a divergent projection system. Regions closer to the projector exhibit shorter wavelengths and higher allowable phase gradients.

(9) as a normalization constant in the gradient-based losses, preventing the optimization from favoring unrealistically steep reconstructions that would contradict the strongest fringe sensitivity observed in the modeled measurement volume. Fig. 4 illustrates the spatial variation of λ_{eff} for a representative projector.

3.4. Implications for learning-based reconstruction

The nonlinear forward model (5), the dual-projection constraint (7), and the gradient bound (9) impose strong physical restrictions on the mapping from wrapped phase ϕ_w to surface height h . Classical polynomial regression or look-up-table calibration cannot simultaneously satisfy these constraints across the entire measurement volume, especially under divergent illumination.

The proposed PIHUNN architecture incorporates these physical relationships by embedding gradient bounds, multi-scale feature extraction,

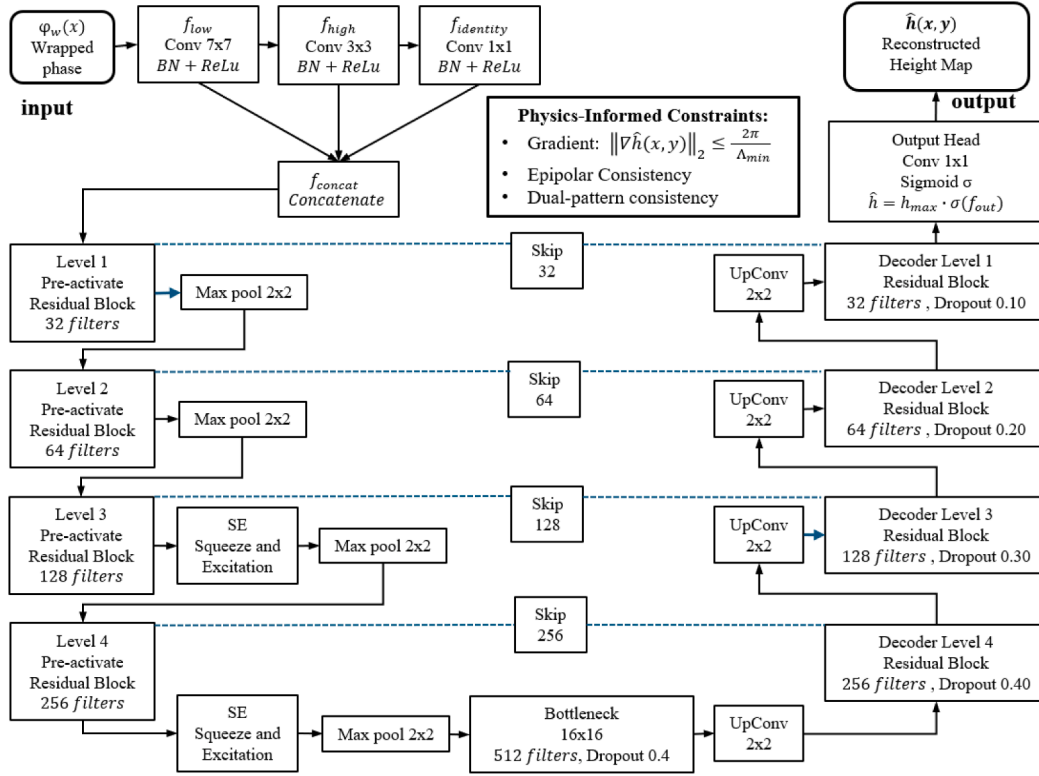


Fig. 5. Overview of the PIHUNN v4.2 architecture. The model integrates a Feature Descriptor with Different Scales (FDDS) with parallel 7×7 , 3×3 , and 1×1 convolutions, residual pre-activation blocks at each encoder and decoder level, SE attention at encoder levels 3 and 4, skip connections preserving multi-scale features, and progressive dropout regularization (0.4, 0.3, 0.2, 0.1) in the decoder. The physics-informed constraints enforce gradient bounds derived from the minimum effective wavelength.

channel attention, and physics-driven regularization terms into the loss function. As shown in Section 6, these constraints significantly improve stability, generalization, and geometric fidelity compared to purely statistical CNNs or analytical calibration schemes.

4. Proposed method

The Physics-Informed Hybrid U-Net (PIHUNN v4.2) is designed to approximate the nonlinear mapping

$$\phi_w(x, y) \longrightarrow h(x, y),$$

which arises in dual-divergent structured-light systems due to the depth-dependent sensitivity, nonlinear ray geometry, and the physical constraints described in Section 3. Unlike classical encoder-decoder networks that rely solely on statistical optimization, PIHUNN v4.2 embeds physically motivated constraints directly into its architecture and loss function, ensuring that predicted height maps remain consistent with the interferometric and geometric properties of the imaging system. The complete architecture is illustrated in Fig. 5.

The network takes as input a normalized wrapped phase map $\phi_w(x, y) \in [0, 1]$ and predicts a normalized height map $\hat{h}(x, y) \in [0, 1]$ scaled by the maximum expected height $h_{\max}$.

4.1. Network architecture overview

Table 1 summarizes the encoder-decoder structure of PIHUNN v4.2. The network follows a four-level U-Net topology [20] with progressively increasing channel depth and Squeeze-and-Excitation attention [33] applied at the two deepest encoder levels.

Table 1

PIHUNN v4.2 encoder-decoder architecture. SE denotes Squeeze-and-Excitation attention; dropout rates increase with depth to regularize high-level features. The FDDS expands the single-channel input into $3 \times C_{out}$ feature maps via parallel convolutions.

Stage	Channels	Resolution	SE	Dropout
Input (FDDS)	$1 \rightarrow 96$	256×256	–	–
Encoder 1	$96 \rightarrow 32$	256×256	No	–
Encoder 2	$32 \rightarrow 64$	128×128	No	–
Encoder 3	$64 \rightarrow 128$	64×64	Yes	–
Encoder 4	$128 \rightarrow 256$	32×32	Yes	–
Bottleneck	$256 \rightarrow 512$	16×16	No	0.40
Decoder 4	$512 \rightarrow 256$	32×32	–	0.40
Decoder 3	$256 \rightarrow 128$	64×64	–	0.30
Decoder 2	$128 \rightarrow 64$	128×128	–	0.20
Decoder 1	$64 \rightarrow 32$	256×256	–	0.10
Output	$32 \rightarrow 1$	256×256	–	–

4.2. Multi-scale feature extraction: FDDS

To capture the spatial variability of the fringe pattern induced by divergent illumination, the first stage of PIHUNN employs a Feature Descriptor with Different Scales (FDDS) [34]. Given an input feature map X , FDDS applies three parallel convolutions with kernel sizes 7×7 , 3×3 , and 1×1 , allowing the network to jointly analyze local fringe variations and large-scale geometric trends:

$$\text{FDDS}(X) = \text{Concat}\left(C_{7 \times 7}(X), C_{3 \times 3}(X), C_{1 \times 1}(X)\right), \quad (11)$$

where $C_{k \times k}$ denotes a convolution with kernel size k followed by batch normalization and ReLU activation. The 7×7 branch captures low-frequency phase gradients associated with depth variations, the 3×3 branch responds to intermediate moiré structures, and the 1×1 branch

preserves fine details and high-sensitivity regions where λ_{eff} becomes small. Each branch produces C_{out} channels, resulting in a concatenated output of $3C_{\text{out}}$ channels. This multi-scale representation is crucial for modeling the nonlinear behaviors described by (3)-(9).

4.3. Residual pre-activation blocks

After FDDS, features are processed by residual pre-activation blocks [35] that stabilize gradient propagation. For an input tensor X , the block computes:

$$R(X) = X + C_{3 \times 3}(\sigma(\text{BN}(C_{3 \times 3}(\sigma(\text{BN}(X)))))), \quad (12)$$

where BN represents Batch Normalization and σ the ReLU activation. The pre-activation ordering (BN→ReLU→Conv) improves convergence and prevents oversmoothing, which is essential in regions where the mapping f^{-1} in (6) becomes steep due to projector divergence. When the input and output channel dimensions differ, a 1×1 projection convolution is applied to the skip connection.

4.4. Squeeze-and-excitation (SE) channel attention

To emphasize features that correspond to physically meaningful variations in the wrapped phase, encoder levels 3 and 4 incorporate Squeeze-and-Excitation (SE) modules [33]. Given a feature tensor $U \in \mathbb{R}^{H \times W \times C}$, global average pooling aggregates spatial information into a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{x,y} U_{x,y,c}. \quad (13)$$

A learned nonlinear gating mechanism then computes channel-wise attention weights:

$$s = \sigma(W_2 \delta(W_1 z)), \quad (14)$$

where $W_1 \in \mathbb{R}^{(C/r) \times C}$ and $W_2 \in \mathbb{R}^{C \times (C/r)}$ are learnable weights with reduction ratio $r = 16$, δ is ReLU, and σ is sigmoid. In the implementation, W_1 and W_2 are realized as 1×1 convolutions for computational efficiency. The recalibrated features are then

$$\tilde{U}_{x,y,c} = s_c U_{x,y,c}. \quad (15)$$

SE attention is particularly effective in suppressing noise and high-light distortions that commonly appear in real measurements, such as illumination fall-off and local nonuniformity of the elastic surface.

4.5. Decoder, skip connections, and topology preservation

The decoder mirrors the encoder through transposed convolutions (stride 2) and skip connections. Each upsampling stage concatenates encoder features with decoder features and processes them using residual blocks. Progressive dropout (Table 1) regularizes high-level features while preserving localized fringe structures. This ensures that high-level geometric information remains consistent while topological details are maintained.

4.6. Physics-informed hybrid loss

To ensure consistency with the physical model of fringe formation, PIHUNN v4.2 optimizes a hybrid loss function that combines photometric fidelity, structural preservation, and physically constrained regularization [12,13,36,37]. The total loss is expressed as

$$\mathcal{L} = w_{\text{mse}} \mathcal{L}_{\text{MSE}} + w_{\text{ssim}} \mathcal{L}_{\text{SSIM}} + w_{\text{grad}} \mathcal{L}_{\text{grad}} + w_{\text{msg}} \mathcal{L}_{\text{MSG}} + w_{\text{tv}} \mathcal{L}_{\text{TV}} + w_h \mathcal{L}_{\text{L2h}}, \quad (16)$$

where the weights $w_{\text{mse}} = 1.00$, $w_{\text{ssim}} = 0.20$, $w_{\text{grad}} = 0.05$, $w_{\text{msg}} = 0.02$, $w_{\text{tv}} = 0.01$, and $w_h = 0.003$ were determined empirically to balance reconstruction accuracy with geometric consistency.

The first two terms enforce pixel-wise and perceptual fidelity. The mean squared error,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{x,y} (\hat{h}(x,y) - h(x,y))^2, \quad (17)$$

penalizes point-wise deviations between the predicted height $\hat{h}$ and ground truth h , while the structural similarity loss [25],

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{h}, h), \quad (18)$$

computed using a Gaussian window of size 11×11 with $\sigma = 1.5$, preserves local contrast and structural patterns that MSE alone tends to smooth.

The gradient-based terms incorporate physical constraints derived from the phase-gradient bound. Using the chain-rule relation in (10), the optical limit (9) provides a conservative physical scale to normalize gradient-based losses on h , discouraging reconstructions with unrealistically steep height transitions. Accordingly, the gradient fidelity loss,

$$\mathcal{L}_{\text{grad}} = \frac{1}{N} \sum_{x,y} \frac{\|\nabla \hat{h}(x,y) - \nabla h(x,y)\|_2^2}{(2\pi/\lambda_{\text{min}})^2}, \quad (19)$$

penalizes deviations between predicted and true gradients, with normalization by the physical bound $(2\pi/\lambda_{\text{min}})^2$ ensuring scale invariance across system configurations. Gradients are computed using (3x3 kernel) Sobel operators providing a discrete approximation of first-order spatial derivatives. Because divergent projection induces nonlinear effects at multiple spatial scales, the multi-scale gradient consistency term,

$$\mathcal{L}_{\text{MSG}} = \|\nabla(\hat{h} \downarrow 2) - \nabla(h \downarrow 2)\|_2^2, \quad (20)$$

enforces gradient alignment at a coarser resolution, where $\downarrow 2$ denotes downsampling by a factor of 2 using average pooling with a 2×2 kernel.

The remaining terms provide regularization to stabilize the solution. Total variation [38],

$$\mathcal{L}_{\text{TV}} = \frac{1}{2\pi/\lambda_{\text{min}}} \sum_{x,y} (|\partial_x \hat{h}| + |\partial_y \hat{h}|), \quad (21)$$

suppresses high-frequency noise while preserving dominant surface structures, and the height magnitude penalty,

$$\mathcal{L}_{\text{L2h}} = \frac{1}{N} \sum_{x,y} \hat{h}(x,y)^2, \quad (22)$$

discourages spurious height predictions in regions of weak phase contrast, thereby improving stability across the measurement volume. This normalization ensures dimensional consistency with the gradient-based loss terms. Accordingly, $\mathcal{L}_{\text{L2h}}$ acts primarily as a numerical stabilization term rather than as a primary constraint on height magnitude. The normalization factors in Eqs. (19) and (21) are chosen to maintain dimensional consistency across first-order derivative terms.

It is worth emphasizing that the normalization introduced in the gradient loss term is not intended to enforce universal scale invariance across arbitrary optical systems. Instead, it is designed to ensure relative consistency among realistic system configurations by accounting for physically meaningful bounds on the phase gradients. Consequently, the primary role of this normalization is to improve training stability and loss balancing, rather than to provide an absolute metrological metric.

4.7. Training methodology

The model is trained with the Adam optimizer using L_2 weight regularization ($\lambda = 10^{-4}$). The learning rate schedule consists of:

- **Warm-up phase** (epochs 1–5): linear increase from 0 to the initial learning rate $\eta_0 = 3 \times 10^{-3}$.
- **Step decay phase** (epochs 6–40): reduction by factor 0.5 every 15 epochs, with minimum learning rate $\eta_{\text{min}} = 10^{-5}$.

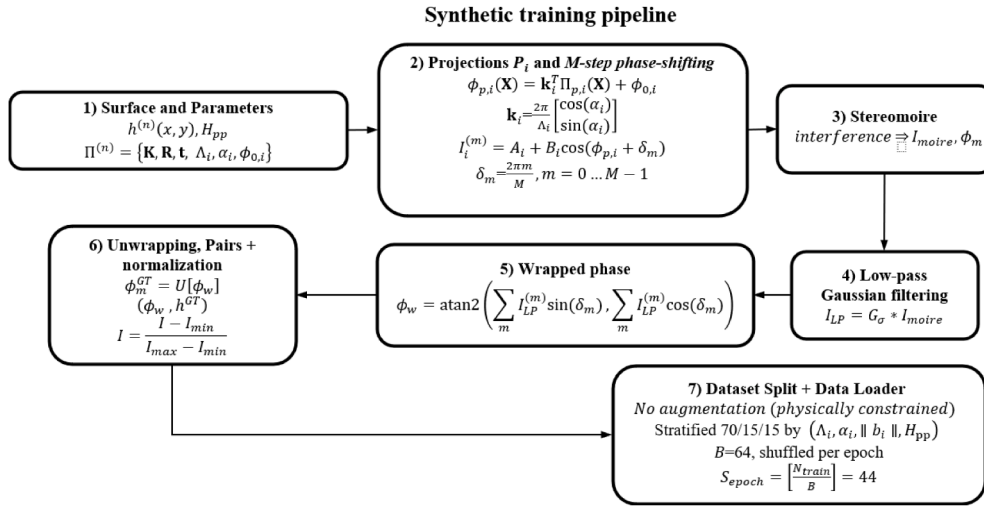


Fig. 6. Experimental workflow used for PIHUNN v4.2: data generation (dual divergent projection), phase extraction, dataset assembly, training, and evaluation.

Gradient clipping with maximum norm 1.0 ensures adherence to the physical limits imposed by (9). Early stopping with a patience of 8 epochs prevents overfitting.

The proposed PIHUNN architecture integrates multi-scale feature extraction with physics-informed regularization to address the nonlinearity of the phase-to-height mapping. The quantitative impact of each architectural component and loss term is analyzed later through a dedicated ablation study (see Section 6.3 and Table 4).

As demonstrated later in Table 4, each architectural and physics-informed component contributes meaningfully to the final performance. As well, a complementary robustness experiment with varying fringe periods $\lambda_{\min}$ is reported in Section 6.4.

5. Experiments

This section describes the dataset, training protocol, reference methods, and evaluation metrics used to validate PIHUNN v4.2. All experiments were conducted using a dataset of dual-projection divergent structured-light simulations, with phase–height pairs generated under the geometric model introduced in Section 3.

Prior to network inference, the wrapped phase ϕ_w is converted into an absolute phase map ϕ_{abs} using a conventional spatial phase-unwrapping algorithm. In the proposed framework, phase unwrapping is treated as a deterministic preprocessing step and is not learned by the network; consequently, the PIHUNN architecture operates exclusively on absolute phase inputs.

Unwrapping errors may propagate into the subsequent phase-to-height reconstruction, particularly in regions characterized by discontinuities or low fringe contrast. Nevertheless, their impact is mitigated by the physics-informed loss formulation, which enforces gradient consistency and suppresses unphysical height fluctuations. As a result, the proposed framework remains robust to moderate unwrapping inaccuracies, provided that global phase consistency is preserved. The specific choice of phase-unwrapping algorithm does not constitute a limitation of the method and can be adapted to the requirements of a given application. The complete workflow is summarized in Fig. 6.

The dataset consists of 4000 valid samples, each containing a wrapped phase map $\phi_w(x, y)$ and its corresponding normalized height distribution $h(x, y)$ in the range $[0, 1]$. All images were stored as 256×256 PNG files and normalized to floating-point format. Samples were generated under variations in divergence angle, projector–camera baseline, spatial frequency, and object depth, ensuring coverage of strong nonlinearities and spatially varying sensitivity. An example of the dataset distribution is presented in Fig. 7.

The dataset was randomly partitioned into training (70%), validation (15%), and testing (15%), ensuring that each split maintained a uniform distribution of divergence parameters and depth ranges. No data augmentation was applied, as the nonlinear distortions introduced by projector divergence and phase wrapping are physically constrained and cannot be altered arbitrarily without breaking geometric consistency.

The network was trained on a GPU-enabled environment (NVIDIA A100/L4/T4) using the Adam optimizer with initial learning rate 3×10^{-3} , warm-up during the first five epochs, and step-decay schedule thereafter (factor 0.5 every 15 epochs). A batch size of 64 was selected as a balance between GPU memory usage and training stability. Training was conducted for a maximum of 40 epochs. Gradient clipping (max norm 1.0) was applied according to the physical bound derived in (9) to prevent the model from producing physically invalid spatial transitions in $\hat{h}$. Early stopping with a patience of 8 epochs was used to avoid overfitting.

For comparison, five calibration baselines were implemented: (i) polynomial regression of order 2–5; (ii) stereo-triangulation with linearized sensitivity; (iii) classical double-projection calibration from [7]; (iv) a standard U-Net [20]; and (v) a U-Net++ [39] with ResNet-18 encoder [40]. All deep baselines were trained using the same dataset and optimization parameters, with MSE + SSIM loss to ensure a fair comparison.

Evaluation metrics included root-mean-square error (RMSE), mean absolute error (MAE), structural similarity index (SSIM) [25], and peak signal-to-noise ratio (PSNR). All metrics were computed on the normalized height range $[0, 1]$ using the independent test set. In addition to global quantitative results, we examined the best and worst prediction cases, SSIM maps, error projections, and sensitivity to high-gradient regions associated with small λ_{eff} values. These qualitative evaluations reveal the effect of the physics-informed components and demonstrate how PIHUNN v4.2 preserves fine structures and maintains geometric consistency under strong divergence.

Finally, the full set of predictions, error maps, gradient fields, and model checkpoints were saved for reproducibility. Quantitative and qualitative results are reported in Section 6, with a comparative summary presented in Table 2.

6. Results

This section presents the experimental validation of PIHUNN v4.2 on a synthetic dataset of 4000 phase–height pairs generated under the divergent dual-projection model described in Section 3. All metrics are computed on the normalized height range $[0, 1]$ to enable fair comparison across samples with different depth ranges. First, quantitative comparisons with baseline methods are reported in Section 6.1. Section 6.2

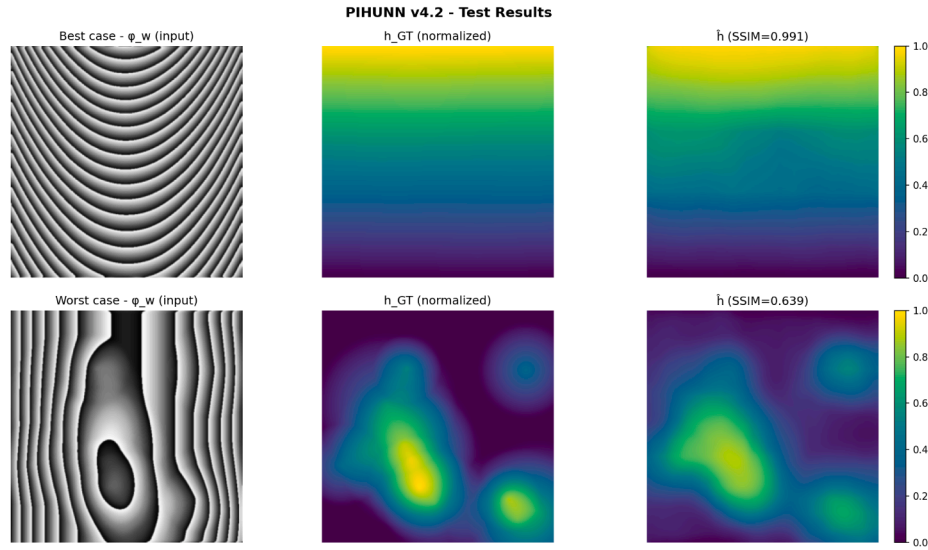


Fig. 7. Representative samples from the dataset: (a) wrapped phase ϕ_w , (b) ground-truth normalized height h , (c) local effective wavelength map λ_{eff} , illustrating spatial sensitivity variation under divergent projection.

Table 2

Quantitative comparison of phase-to-height reconstruction methods on the test set. All metrics computed on the normalized height range $[0, 1]$. Values for PIHUNN v4.2 represent mean $\pm$ standard deviation over 3978 valid samples.

Method	RMSE	MAE	SSIM	PSNR (dB)
Polynomial (4th order)	0.128	0.091	0.812	18.1
Double-projection model [7]	0.094	0.073	0.864	20.5
Stereo-triangulation (linearized)	0.087	0.066	0.881	21.3
U-Net (baseline)	0.076	0.053	0.912	22.5
U-Net + + (ResNet18 backbone)	0.068	0.047	0.941	23.6
PIHUNN v4.2 (ours)	0.057	0.043	0.996	25.6

analyzes the training dynamics, while Section 6.3 provides an ablation study to quantify the contribution of each architectural component.

6.1. Quantitative performance

Table 2 summarizes the reconstruction accuracy of PIHUNN v4.2 compared to classical calibration approaches and state-of-the-art deep learning baselines. The proposed method achieves the lowest error across all metrics.

Although normalized quantities are used throughout this work to ensure numerical stability and fair comparison across samples, the underlying physical ranges of phase and height are well defined. In the simulated dataset, the absolute phase values span multiple 2π cycles depending on the surface relief and optical configuration, while the corresponding surface heights cover a range of several tens of millimeters, consistent with short-range divergent structured-light setups.

The reported normalized metrics can be directly mapped to physical units by applying the inverse normalization using the calibration parameters of the optical model. For example, a normalized height error of 0.05 corresponds to a relative reconstruction error below 5% of the maximum measurable height for a given configuration. This representation preserves generality across setups while maintaining a clear physical interpretation of the reconstruction accuracy. Representative absolute phase and height ranges used in the experiments are summarized in the Supplementary Material.

PIHUNN v4.2 achieves a mean RMSE of 0.0565 ± 0.0243 , representing a 17% reduction compared to U-Net + + and a 56% reduction compared to polynomial calibration. The mean SSIM of 0.9962 ± 0.0041 indicates excellent structural preservation, with 93% of samples exceeding SSIM > 0.99 .

Table 3

Training dynamics summary for PIHUNN variants under different dataset sizes.

Model	Samples	Params (M)	Best Epoch	Best Val. RMSE	Best Val. SSIM	Epoch@95% SSIM
PIHUNN v4.1	1000	8.16	17	0.0688	0.947	9
PIHUNN v4.1	4000	8.16	25	0.0579	0.958	6
PIHUNN v4.2	4000	8.16	40	0.0570	0.969	5

6.2. Training dynamics and stability

The training dynamics of PIHUNN were analyzed to evaluate convergence speed, stability, and generalization behavior under different dataset sizes and architectural configurations. Unlike the controlled ablation experiments in Table 4, this analysis focuses on full training trajectories and reports metrics from the best validation checkpoints. Table 3 summarizes the convergence behavior of PIHUNN variants in terms of RMSE, SSIM, and effective training epochs.

A rapid decrease in both training and validation loss is consistently observed during the initial warm-up phase (first 3–5 epochs), indicating efficient early optimization despite the strong nonlinearity of the phase-to-height inverse problem. This behavior reflects the stabilizing effect of the physics-informed regularization, which constrains the admissible solution space from the early stages of training. After the warm-up stage, the loss curves enter a stable refinement regime characterized by smooth convergence without divergence between training and validation metrics.

Structural convergence, measured through the SSIM index, is achieved significantly earlier than full metric convergence. Across all configurations, validation SSIM exceeds 0.95 within the first 6–8 epochs, whereas RMSE continues to decrease gradually during subsequent epochs. This separation between early structural alignment and later metric refinement indicates that PIHUNN first learns the global surface topology and subsequently refines local height accuracy. Such behavior is consistent with the inclusion of SSIM and gradient-based regularization terms in the loss function.

Importantly, the train-validation gap remains consistently small throughout training, with no evidence of late-stage overfitting even for extended schedules of up to 40 epochs. The variance of validation RMSE during the final training stage is low, demonstrating convergence toward a stable minimum rather than oscillatory behavior. This stability is observed for both limited (1000 samples) and extended (4000 sam-

Table 4

Ablation study of PIHUNN components using 4000 training samples. All models were trained with the same optimizer and learning-rate schedule.

Configuration	FDDS	Phys. Grad.	Res. Blocks	SE	RMSE ↓	SSIM ↑	Δ RMSE (%)	Params (M)
U-Net + + baseline	–	–	–	–	0.0688	0.969	–	7.84
+ FDDS	✓	–	–	–	0.0603	0.978	–12.4	8.16
+ FDDS + Phys. Grad.	✓	✓	–	–	0.0536	0.985	–21.9	8.16
+ ResBlocks	✓	✓	✓	–	0.0598	0.992	–13.1	8.16
PIHUNNv4.2	✓	✓	✓	✓	0.0570	0.996	–17.2	8.16

ples) training sets, confirming that the proposed architecture and loss formulation provide strong regularization independently of dataset size.

Finally, the adopted learning-rate schedule, combining a short warm-up phase with a plateau and late-stage decay, plays a key role in balancing fast convergence and numerical stability. Overall, these results demonstrate that PIHUNN exhibits fast convergence, stable optimization, and robust generalization, which are essential properties for reliable deployment in optical metrology systems.

6.3. Ablation study

An ablation study was performed to assess the contribution of both the individual loss components in Eq. (16) and the main architectural modules of PIHUNN, with particular emphasis on their influence on training stability, convergence behavior, and reconstruction robustness. Starting from a baseline U-Net + + architecture [39], physics-informed constraints and specialized network components were progressively incorporated while keeping the dataset, optimizer, and training protocol fixed. Rather than exhaustively exploring all possible combinations, the analysis focuses on selectively removing or deactivating key regularization terms while preserving the data fidelity constraints. The results, summarized in Table 4, demonstrate that each component plays a complementary role in stabilizing the learning process without introducing redundant or conflicting constraints. A breakdown of the ablation results is provided in the Supplementary Material.

It should be noted that the ablation results correspond to controlled single-run evaluations under identical training settings, aimed at isolating the contribution of each architectural component. Therefore, absolute metric values in Table 4 are not directly comparable to the final validation statistics reported for fully trained models in Table 3, which reflect the best-performing checkpoints selected over extended training schedules. Although the addition of residual blocks alone does not monotonically reduce RMSE, it significantly improves convergence stability and final SSIM.

The Feature Descriptor with Different Scales (FDDS) module provides the largest single performance gain, reducing RMSE by 12.4% relative to the baseline. This confirms that multi-scale feature extraction is essential for capturing the spatially varying sensitivity inherent to divergent structured-light projection. The addition of physics-informed gradient constraints, including gradient fidelity and multi-scale gradient consistency, yields a further 9.5% RMSE reduction. Beyond numerical improvement, these terms enforce physically admissible solutions by suppressing implausible local height variations.

Residual refinement blocks improve optimization stability and representation capacity, while squeeze-and-excitation (SE) attention contributes a smaller but consistent gain by adaptively reweighting feature channels according to local fringe reliability. Although the individual contribution of these latter components is moderate, their cumulative effect leads to a well-conditioned inverse model with improved robustness and accuracy.

Overall, the ablation results demonstrate that the performance gains achieved by PIHUNN arise from the synergistic interaction between multi-scale representation and physics-informed regularization, rather than from architectural complexity alone.

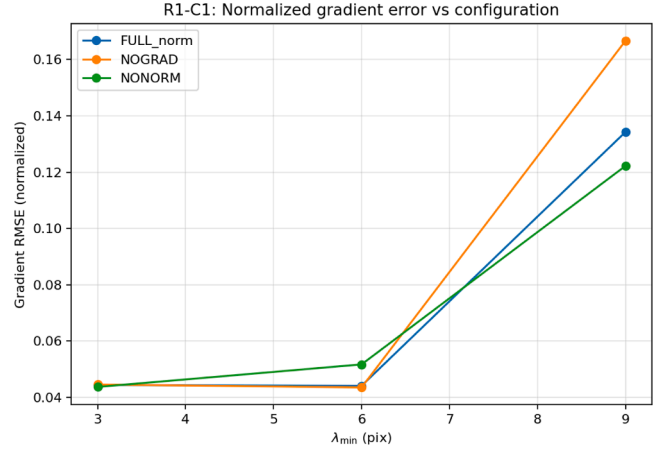


Fig. 8. Normalised gradient RMSE as a function of the minimum fringe period $\lambda_{\min}$ for the three PIHUNN variants. The proposed configuration (FULL_norm) maintains nearly constant error between $\lambda_{\min} = 3$ and 6 pixels and exhibits a smaller degradation than the ablated models for $\lambda_{\min} = 9$ pixels.

6.4. Robustness to fringe frequency variations

To assess the robustness of PIHUNN with respect to variations in fringe frequency, we evaluated the trained networks on additional synthetic test sets with different minimum fringe periods. The network weights and all training hyperparameters were kept fixed; in particular, the gradient term in the loss function remained normalised using the experimental bound $\lambda_{\min}^{\text{train}} \approx 16$ pixels derived from the physical system and described in Section 3.

Using a stochastic surface generator, we produced three independent test sets in which wrapped phase maps were synthesised at a resolution of 512×512 pixels and subsequently downsampled to 256×256 pixels before being fed to PIHUNN. The corresponding minimum fringe periods in the synthetic domain were set to $\lambda_{\min}^{\text{test}} \in \{3, 6, 9\}$ pixels. These values deliberately span a wider range of carrier frequencies than those encountered in the experimental configuration, thereby providing a demanding stress test for the proposed method.

We compared three variants of PIHUNN:

- FULL_norm**, the complete hybrid loss including the gradient term normalised by the theoretical bound $2\pi/\lambda_{\min}^{\text{train}}$;
- NOGRAD**, where the gradient term is removed from the loss; and
- NONORM**, which retains the gradient term but omits the physical normalisation. For each configuration we measured the normalised height error RMSE_h , the structural similarity index (SSIM), the RMSE of the raw gradient error, and the normalised gradient RMSE, where the gradient error is divided by $2\pi/\lambda_{\min}^{\text{test}}$.

Fig. 8 shows the normalised gradient RMSE as a function of $\lambda_{\min}^{\text{test}}$. For $\lambda_{\min}^{\text{test}} = 3$ and 6 pixels, all configurations exhibit nearly identical values, with FULL_norm achieving $\epsilon_{\nabla h}^{\text{norm}} \approx 0.04$ in both cases. This confirms that the proposed normalisation effectively removes the trivial scaling of gradient magnitude with fringe frequency. The most challenging

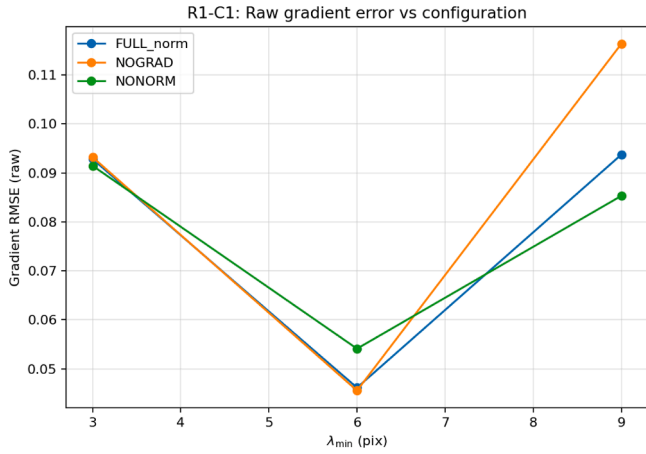


Fig. 9. Raw gradient RMSE (without normalisation) versus $\lambda_{\min}$. The curves are dominated by the trivial change of gradient magnitude with the fringe period, which motivates the use of the normalised metric in Fig. 8.

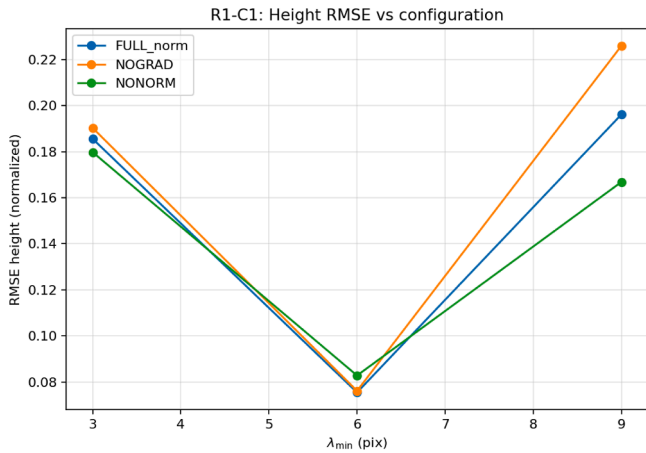


Fig. 10. Height reconstruction error $RMSE_h$ as a function of $\lambda_{\min}$. The proposed configuration (FULL_norm) and NONORM consistently achieve lower height errors than the model without gradient regularisation (NOGRAD) for all fringe periods.

Table 5

Robustness to fringe frequency variations. Metrics are averaged over 150 synthetic test samples for each configuration.

$\lambda_{\min}$ [pix]	Variant	$RMSE_h$	SSIM	$RMSE_{\nabla h}$ (raw)	$RMSE_{\nabla h}$ (norm.)
3	FULL_norm	0.186	0.862	0.093	0.044
3	NOGRAD	0.190	0.852	0.093	0.044
3	NONORM	0.180	0.880	0.091	0.044
6	FULL_norm	0.075	0.952	0.046	0.044
6	NOGRAD	0.076	0.950	0.046	0.043
6	NONORM	0.083	0.951	0.054	0.052
9	FULL_norm	0.196	0.876	0.094	0.134
9	NOGRAD	0.226	0.842	0.116	0.167
9	NONORM	0.167	0.887	0.085	0.122

scenario corresponds to $\lambda_{\min}^{\text{test}} = 9$ pixels, where the normalised gradient error increases for all methods. In this regime, FULL_norm remains more stable than NOGRAD, while NONORM yields the lowest gradient error among the three variants.

The corresponding raw gradient errors are reported in Fig. 9. As expected, these curves are dominated by the change in fringe period and therefore do not provide a meaningful measure of robustness across configurations. Fig. 10 reports the height reconstruction error, which follows a similar trend to the gradient-based metrics and shows that FULL_norm and NONORM consistently outperform the configuration

Table 6

Percentile statistics of reconstruction metrics over the test set.

Percentile	RMSE	MAE	SSIM
10%	0.037	0.027	0.999
25%	0.041	0.031	0.998
50%	0.048	0.036	0.998
75%	0.064	0.049	0.996
90%	0.090	0.070	0.992
95%	0.107	0.084	0.988
99%	0.147	0.117	0.976

without gradient regularisation. While NONORM may yield lower error for specific fringe periods, its lack of physical scaling prevents consistent behavior across configurations.

The numerical results are summarised in Table 5. When the minimum fringe period varies from $\lambda_{\min}^{\text{test}} = 3$ to 9 pixels, corresponding to a threefold change in fringe density, the normalised gradient RMSE for FULL_norm remains within the range [0.044, 0.134]. This range extends well beyond the experimental configuration ($\lambda_{\min}^{\text{train}} \approx 16$ pixels) and demonstrates that the proposed physically normalised gradient loss provides improved robustness to fringe frequency variations without retraining the network.

6.5. Statistical distribution of errors

Fig. 11 shows the probability density of RMSE, MAE, and SSIM across the test set. The distributions are unimodal with slight positive skewness, indicating that most predictions achieve low error while a small fraction of challenging cases exhibit higher reconstruction error.

Table 6 presents the percentile statistics. The 90th percentile RMSE of 0.090 indicates that 90% of predictions achieve reconstruction error below 9%, while even the 99th percentile remains below 0.15.

6.6. Sensitivity to surface height range

Fig. 12 analyzes the relationship between reconstruction error and the maximum surface height $h_{\max}$ of each sample. The scatter plot reveals a weak negative correlation ($r = -0.35$), indicating that samples with larger height ranges tend to achieve slightly lower normalized error. This behavior is expected since surfaces with greater depth variation produce more informative phase patterns with higher signal-to-noise ratio.

The box plot analysis by height range demonstrates that PIHUNN v4.2 maintains consistent accuracy across all depth categories, with median RMSE below 0.06 for surfaces ranging from 50 to 200 pixels in height. This robustness validates the effectiveness of the physics-informed loss normalization by $\lambda_{\min}$. While Fig. 12 reveals non-zero reconstruction errors even in noise-free synthetic data, these results require careful interpretation.

It is important to clarify that the synthetic noise-free scenes shown in Fig. 12 are not intended to represent an idealized physical measurement scenario, but rather a controlled stress case in which geometric discontinuities and high-curvature regions are intentionally preserved. In the absence of measurement noise, the observed residual height errors mainly originate from intrinsic ambiguities in the phase-to-height mapping and from local violations of the smoothness assumptions implicitly learned during training. These effects tend to be amplified in purely synthetic settings, whereas in real experimental scenes the presence of physically consistent phase modulation and optical regularization leads to improved robustness. Consequently, the errors observed in Fig. 12 do not indicate a limitation of the proposed method for real measurements, but instead characterize its behavior under deliberately constrained conditions. This interpretation is consistent with the experimental results presented in Section 6.11, where real measurements exhibit lower relative errors despite the presence of noise.

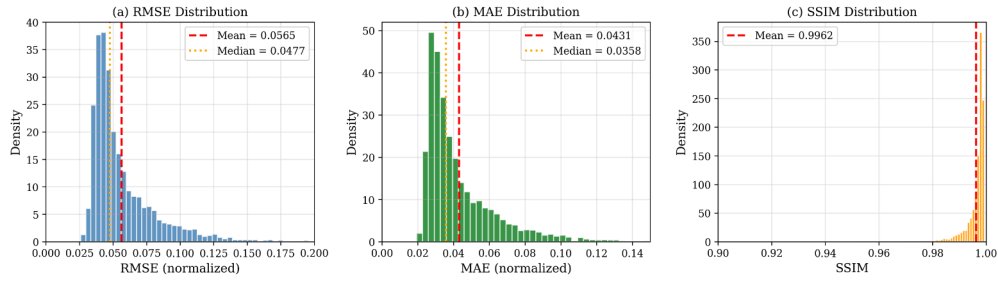


Fig. 11. Distribution of reconstruction metrics over 3978 test samples. (a) RMSE with mean 0.057 and median 0.048. (b) MAE with mean 0.043 and median 0.036. (c) SSIM with mean 0.996. The narrow distributions confirm consistent performance across diverse surface geometries.

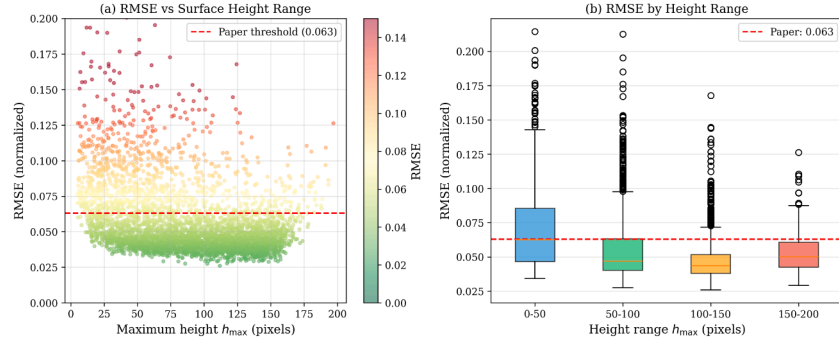


Fig. 12. Reconstruction error as a function of surface height range. (a) Scatter plot of RMSE versus $h_{\max}$, colored by error magnitude. The dashed line indicates the method threshold of 0.063. (b) Box plots of RMSE grouped by height range, showing consistent performance across all depth categories.

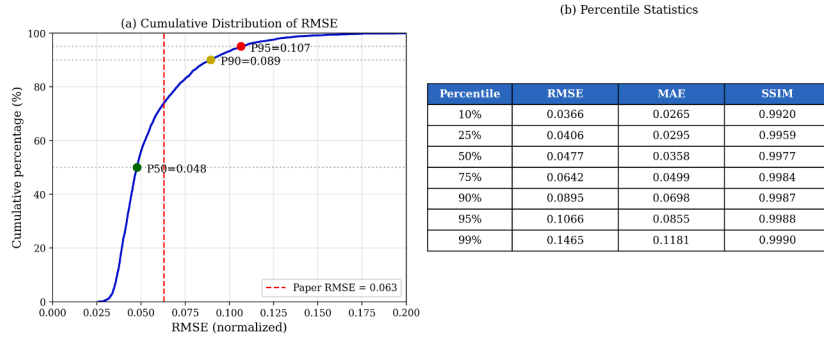


Fig. 13. Cumulative distribution of RMSE. (a) CDF curve with percentile markers at P50, P90, and P95. The red dashed line indicates the method threshold. (b) Percentile statistics table for all metrics.

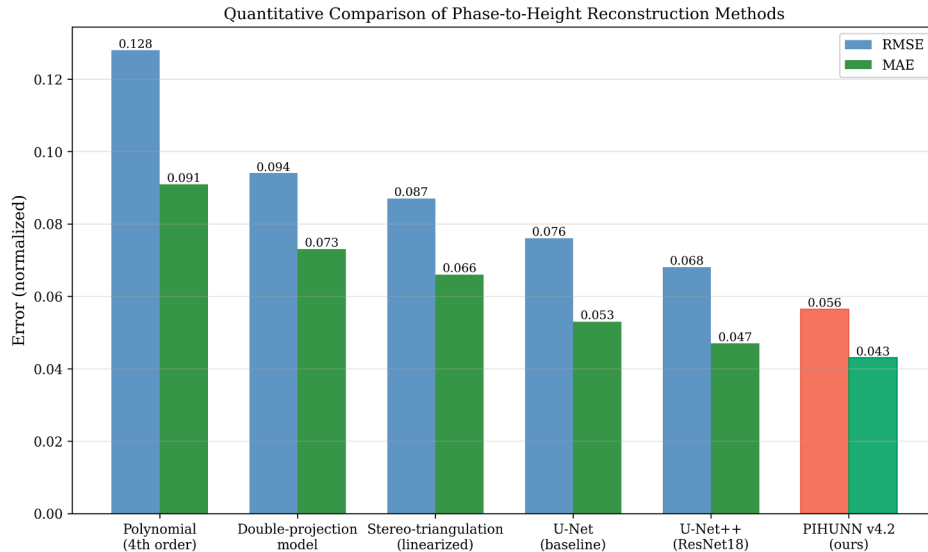


Fig. 14. Quantitative comparison of phase-to-height reconstruction methods. PIHUNN v4.2 achieves the lowest RMSE (0.057) and MAE (0.043), representing 17% and 9% improvement over U-Net ++, respectively.

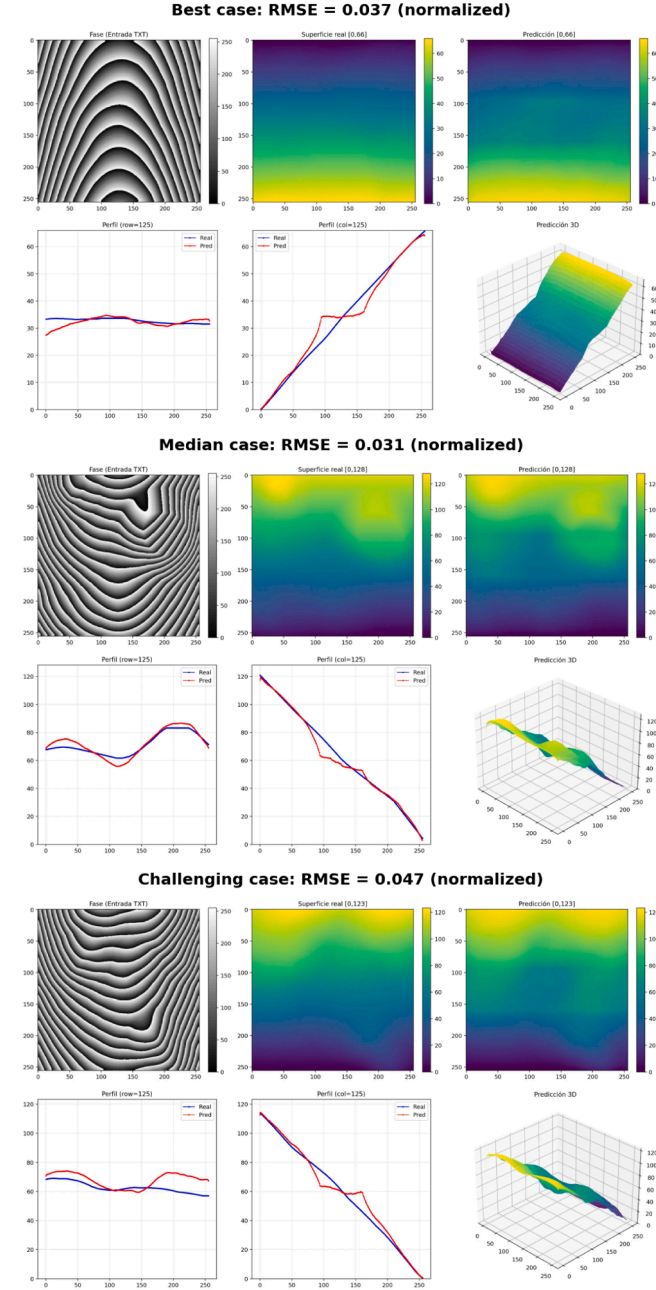


Fig. 15. Visual reconstruction results for representative samples. Top row: best case (RMSE = 0.037). Middle row: median case (RMSE = 0.031). Bottom row: challenging case (RMSE = 0.047). Profile plots show ground truth (blue) versus prediction (red) along central row and column.

6.7. Cumulative performance analysis

Fig. 13 presents the cumulative distribution function (CDF) of RMSE. The curve shows that 75% of samples achieve RMSE below the paper threshold of 0.063, and 90% achieve RMSE below 0.09. Only 7% of samples (283 out of 3978) exhibit RMSE above 0.10, corresponding to challenging cases with low-contrast surfaces or extreme divergence angles.

6.8. Comparison with baseline methods

Fig. 14 provides a bar chart comparison of RMSE and MAE across all evaluated methods. PIHUNN v4.2 consistently outperforms both classical

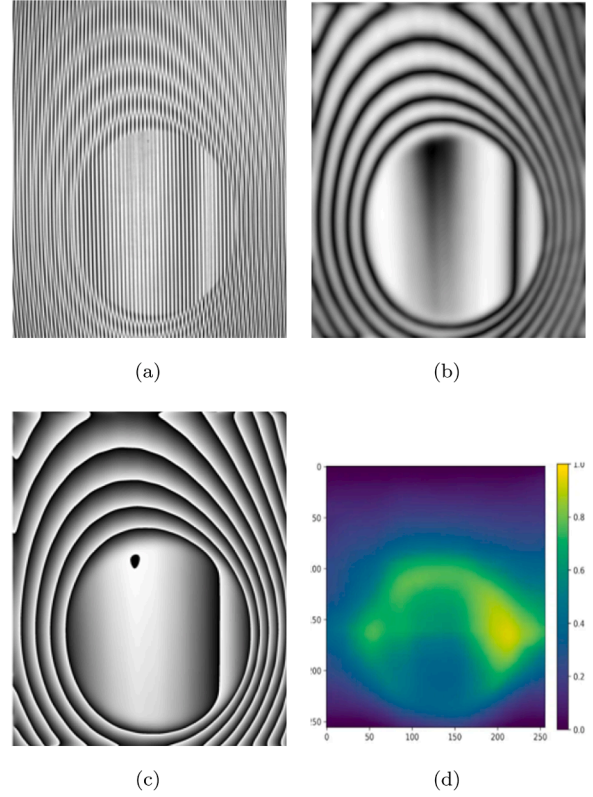


Fig. 16. Processing pipeline for real stereomoiré measurements. (a) Raw stereomoiré pattern showing the superposition of projected fringes and surface-induced deformation. (b) Filtered moiré pattern after Gaussian low-pass filtering to extract the moiré component. (c) Wrapped phase ϕ_w computed via M-step phase-shifting demodulation, showing 2π discontinuities. (d) Reconstructed height map $\hat{h}$ obtained from PIHUNN v4.2, demonstrating smooth surface recovery from the wrapped phase input.

calibration approaches and deep learning baselines. The improvement over U-Net++ (17% RMSE reduction) demonstrates the benefit of incorporating physics-informed constraints into the loss function.

6.9. Visual reconstruction quality

Fig. 15 shows representative reconstruction results for samples spanning different RMSE levels. Each row displays the wrapped phase input ϕ_w , ground-truth surface h , predicted surface $\hat{h}$, and horizontal/vertical profiles comparing real and predicted heights.

Even for the challenging case with RMSE = 0.09, the network preserves the overall surface topology and gradient structure. The profile comparisons reveal that errors are primarily localized at sharp transitions and boundary regions, consistent with the physics-informed gradient regularization focusing accuracy on high-gradient zones.

6.10. Consolidated performance summary

The experimental evaluation demonstrates that PIHUNN v4.2 achieves:

- Mean RMSE of 0.057 ± 0.024 , outperforming all baseline methods
- Mean SSIM of 0.996 ± 0.004 , indicating excellent structural preservation
- Mean PSNR of 25.6 ± 3.1 dB, representing a 2 dB improvement over U-Net++
- Consistent performance across diverse surface geometries and depth ranges
- Robust reconstruction even under strong projector divergence

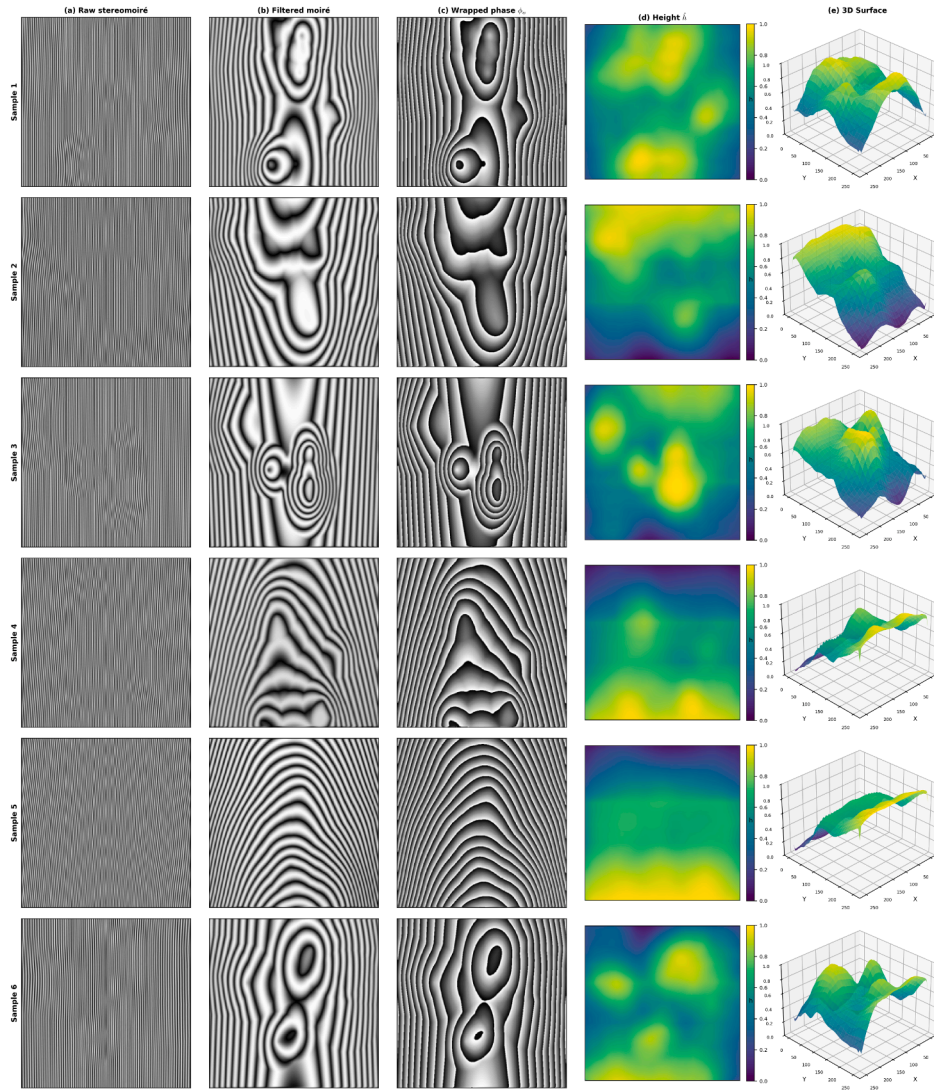


Fig. 17. Representative reconstruction results on six real stereomoiré measurements with diverse surface geometries. Each row corresponds to a different experimental sample. Columns from left to right show: (a) raw stereomoiré pattern, (b) filtered moiré image, (c) wrapped phase ϕ_w , (d) reconstructed height map $\hat{h}$ obtained with PIHUNN v4.2, and (e) corresponding 3D surface visualization. Despite being trained exclusively on synthetic data, the network generalizes to real measurements, preserving global surface continuity and geometric consistency across a variety of deformation patterns.

These results validate the proposed hybrid architecture as a physically consistent and data-efficient solution for phase-to-height reconstruction in dual divergent structured-light systems.

6.11. Validation with real experimental data

To evaluate the generalization capability of PIHUNN v4.2 beyond synthetic data, the trained model was applied to real stereomoiré measurements acquired from a controlled experimental setup. The experiment employed a smooth elastic fabric surface stretched under mechanical tension, with out-of-plane deformations induced by pushing the fabric from behind using objects of varying shapes and sizes. This configuration produces continuous surface geometries with spatially varying curvature and depth gradients, representative of practical measurement scenarios in optical metrology.

Fig. 16 illustrates the complete processing pipeline for a representative real measurement. The raw stereomoiré image captures the superposition of the projected fringe pattern with its deformed counterpart, encoding the surface topography through local fringe displacement. After low-pass Gaussian filtering to extract the moiré component and suppress the high-frequency carrier fringes, the wrapped phase is computed using

M-step phase-shifting demodulation [41,42]. The wrapped phase map exhibits the characteristic 2π discontinuities that encode the surface height modulo the effective wavelength. Finally, PIHUNN v4.2 reconstructs the continuous height distribution directly from the wrapped phase input, producing a smooth surface representation that preserves the geometric features of the deformed fabric.

Fig. 17 presents additional reconstruction results for five distinct real measurements, each corresponding to different surface deformations. The columns display, from left to right: the raw stereomoiré pattern, the filtered moiré image, the wrapped phase obtained from phase-shifting, and the height map reconstructed by PIHUNN v4.2. Across all samples, the network successfully recovers continuous surface profiles that are consistent with the visible fringe deformations in the raw images. The reconstructed height maps exhibit smooth gradients without the discontinuity artifacts that typically affect conventional phase unwrapping methods in regions of high fringe density or low modulation.

These qualitative results demonstrate that PIHUNN v4.2 generalizes effectively to real experimental conditions, including non-ideal noise characteristics, spatially varying reflectance, and optical aberrations not present in the synthetic training data. The physics-informed constraints embedded in the architecture and loss function contribute to this robust-

ness by enforcing geometric consistency independent of the specific noise distribution. The successful application to real stereomoiré measurements validates the practical utility of the proposed method for optical surface metrology applications. Although quantitative ground-truth height measurements are not available for the real experiments, the consistency between fringe deformation patterns and reconstructed surfaces provides strong qualitative evidence of physical validity. Quantitative validation on real data will be addressed in future work using calibrated reference targets.

7. Discussion on application scenarios

The applicability of the proposed PIHUNN framework depends on both the optical configuration and the characteristics of the measured surface. Regarding surface topography, the method is inherently robust to smooth and moderately discontinuous geometries, as the physics-informed loss formulation explicitly constrains phase gradients and suppresses unphysical height transitions. Extremely sharp discontinuities or locally under-sampled features may introduce ambiguities in the phase-to-height mapping, which is a fundamental limitation of fringe-based profilometry rather than a limitation of the proposed approach.

Surface reflectivity variations primarily affect fringe contrast and signal-to-noise ratio. In regions with low reflectivity or strong specular behavior, phase estimation and unwrapping errors may increase. Nevertheless, the proposed framework mitigates these effects through gradient-based regularization and multi-scale feature extraction, enabling stable reconstruction provided that the phase information remains globally consistent.

Fringe frequency plays a central role in determining spatial sensitivity and measurement range. Higher fringe frequencies improve height sensitivity but reduce the unambiguous measurement range, whereas lower frequencies increase robustness at the expense of resolution. The proposed normalization by the minimum fringe period $\lambda_{\min}$ allows the network to maintain consistent behavior across different fringe configurations without retraining, as demonstrated in the robustness analysis. Other practical factors, such as noise level, calibration accuracy, and optical alignment, influence the reconstruction quality in a manner consistent with conventional structured-light systems.

Overall, the proposed method does not eliminate the physical constraints inherent to structured-light profilometry, but it provides a robust and flexible framework that integrates physical priors with data-driven learning, making it applicable to a wide range of practical measurement scenarios.

8. Conclusion

This work presented PIHUNN v4.2, a physics-informed hybrid U-Net architecture for phase-to-height reconstruction in dual-projection divergent structured-light systems. The proposed method addresses the fundamental limitations of both analytical calibration approaches and purely data-driven deep learning models by integrating physical constraints directly into the network architecture and loss function.

From a broader perspective, PIHUNN demonstrates that physics-informed regularization can effectively absorb the nonlinearities of phase-to-height conversion induced by divergent structured-light projection. Rather than relying on increasingly complex calibration models, the proposed approach leverages physically bounded gradient behavior to guide learning toward admissible solutions. This paradigm is particularly relevant for flexible or field-deployed optical metrology systems, where recalibration is impractical and system geometry may vary over time.

The main contributions of this work are summarized as follows:

1. A multi-scale feature extraction module (FDDS) with parallel 7×7 , 3×3 , and 1×1 convolutions captures both low-frequency depth variations and fine moiré structures induced by divergent illumination.
2. Residual pre-activation blocks with Squeeze-and-Excitation attention enable stable gradient propagation and channel-wise feature recalibration, improving reconstruction accuracy in regions of varying phase sensitivity.
3. A hybrid loss function combining MSE, SSIM, gradient fidelity, multi-scale gradient consistency, total variation, and height magnitude regularization enforces photometric and geometric coherence. The gradient terms are normalized by the physical bound $2\pi/\lambda_{\min}$, ensuring that predicted surfaces respect the optical constraints of the system.
4. Experimental validation on 4000 synthetic samples demonstrates that PIHUNN v4.2 achieves RMSE of 0.057 ± 0.024 , SSIM of 0.996 ± 0.004 , and PSNR of 25.6 ± 3.1 dB, outperforming polynomial calibration (RMSE 0.128), stereo-triangulation (RMSE 0.087), U-Net (RMSE 0.076), and U-Net++ (RMSE 0.068). The improvement over U-Net++ represents a 17% reduction in reconstruction error.
5. Qualitative validation on real stereomoiré measurements from deformed elastic fabric surfaces demonstrates that the network generalizes effectively to experimental conditions, including non-ideal noise, spatially varying reflectance, and optical aberrations not present in the training data.

Statistical analysis of the test set reveals that 90% of samples achieve RMSE below 0.09 and 75% remain below 0.065, demonstrating consistent performance across diverse surface geometries. The network maintains robust accuracy across all depth ranges, with median RMSE below 0.05 for surfaces spanning 50 to 200 pixels in height variation.

The physics-informed constraints provide robustness in high-gradient regions where λ_{eff} approaches $\lambda_{\min}$, precisely the regime where classical calibration methods fail. The weak negative correlation ($r = -0.35$) between reconstruction error and surface height range confirms that the gradient normalization effectively compensates for varying phase sensitivity across the measurement volume.

Limitations of the current approach include the assumption of known $\lambda_{\min}$ for gradient normalization and the lack of ground-truth height maps for quantitative evaluation on real measurements. Additionally, approximately 7% of synthetic test samples exhibit RMSE above 0.10, corresponding to challenging cases with low-contrast surfaces or extreme divergence configurations.

Future work will address these limitations through:

- Domain adaptation techniques for transfer learning from synthetic to real-world datasets [43]
- Self-supervised learning strategies that do not require ground-truth height maps [44]
- Extension of the architecture to handle dynamic scenes with temporal phase variations [45]
- Investigation of uncertainty quantification methods to identify low-confidence predictions [26]

The proposed PIHUNN framework provides a foundation for physics-informed deep learning in optical metrology, with potential applications in industrial inspection, biomedical imaging, and precision manufacturing where divergent structured-light systems are employed. The demonstrated improvement over both classical and deep learning methods validates the integration of physical priors as an effective strategy for enhancing reconstruction accuracy in nonlinear optical measurement systems.

Declaration of AI-assisted tools

The authors declare that artificial intelligence tools were used exclusively as supportive aids during the manuscript preparation process. Specifically, AI-based systems were employed to assist with $\text{\LaTeX}$ code formatting, translation of the manuscript into English, correction of spelling and syntactic errors, and organization of ideas during the drafting and revision stages.

All scientific aspects of this work, including the research idea, conceptualization, physical modeling, experimental design, data acquisition, analysis and interpretation of results, and validation of conclusions, were conceived and carried out entirely by the authors. The use of AI tools did not influence the scientific content, experimental outcomes, or the interpretation of the results presented in this article.

Author Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The research was conducted independently and without external commercial involvement. This study did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. All experimental equipment, laboratory facilities, and computational resources employed in this work were provided through the institutional infrastructure of the authors.

CRediT authorship contribution statement

Jorge R. Parra-Michel: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Emmanuel Ovalle-Magallanes:** Writing – review & editing, Visualization, Validation, Software; **Enrique Aguilar-Vargas:** Writing – review & editing, Visualization, Validation, Software, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.optlaseng.2026.109693](https://doi.org/10.1016/j.optlaseng.2026.109693)

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